NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE

Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED | **MATHEMATICS** | **M.M. 81**

REVISION ASSINGMENT # 20 | (SETS, RELATIONS, STATISTICS) | TIME : 75 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. If A, B are two sets then $(A - B) \cup (B - A) \cup (\bar{A} \cap \bar{B})$ is equal to

(A) $\bar{A} \cup \bar{B}$

(B) $\overline{(A \cup B)}$

(C) ϕ

(D) U (where U is universal set and A & B are subsets of U)

2. The first of the two samples in a group has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{1344}$, then the standard deviation of the second sample is :

(A) 8 (B) 6 (C) 4 (D) 5

3. The mean and standard deviation of 20 observations were calculated as 10 and 2.5 respectively. It was found that by mistake one data value was taken as 25 instead of 35. If α and $\sqrt{\beta}$ are the mean and standard deviation respectively for correct data, then (α, β) is :

(A) (11, 26) (B) (10.5, 25) (C) (11, 25) (D) (10.5, 26)

4. Two finite sets A and B have 'm' and 'n' elements respectively. The total number of subsets of A is 56 more then the total number of subsets of B. The value of (m + n) is-

(A) 13 (B) 9 (C) 10 (D) 15

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5. Consider three observations a, b and c such that b = a + c. If the standard deviation of a + 2, b + 2, c + 2 is d, then which of the following is true?

(A) $b^2 = 3(a^2 + c^2) + 9d^2$ (B) $b^2 = a^2 + c^2 + 3d^2$

(C) $b^2 = 3(a^2 + c^2 + d^2)$ (D) $b^2 = 3(a^2 + c^2) - 9d^2$

6. Let in a series of 2n observations, half of them are equal to a and remaining half are equal to –a. Also by adding a constant b in each of these observations, the mean and standard deviation of new set become 5 and 20, respectively. Then the value of $a^2 + b^2$ is equal to :

(A) 425 (B) 650 (C) 250 (D) 925

7. Let M be a relation from R (Set of real number) to R defined by $M = \{(a, b) \mid a, b \in R$ and a + b is an irrational number$\}$, then relation M is -

(A) reflexive and symmetric only (B) symmetric only

(C) symmetric and transitive only (D) equivalence relation

8. Let $S = \{(x, y) | \sin y = \sin x, x, y \in R\}$, then S is

(A) not transitive relation (B) equivalence relation

(C) transitive but not reflexive relation (D) partial order relation

9. Let R be a relation defined on a set of real numbers such that a R b and if $|a| \le b$, then relation R is :-

(A) Reflexive (B) Symmetric (C) Transitive (D) Equivalence

10. $x_i (i = 1,2,3,.....n)$ denotes a distribution whose variance is 10 then variance of a distribution $3x_1 + 2, 3x_2 + 2, 3x_3 + 2, ...3x_n + 2$ will be-

(A) 10 (B) 30 (C) $\frac{10}{3}$ (D) 90

11. For 21 distinct observations, if observations greater than the median are increased by 2, then identify the correct statement-

(A) Mean and median are unchanged. (B) Both median and mean are changed.

(C) Median is changed, mean is same. (D) Median is same, mean is changed.

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12. The arithmetic mean of n observations is $\bar{x}$. If each of these observations is divided by α and increased by β, mean of new series will be -

(A) $\frac{\bar{x}+n\beta}{n}$ (B) $\frac{\bar{x}+\alpha\beta}{\alpha}$ (C) $\frac{\bar{x}+\alpha\beta}{n}$ (D) $\frac{\bar{x}+\alpha\beta}{n\beta}$

13. Let $a_1, a_2, \ldots\ldots a_{101}$ is a group of real numbers such that $a_i > a_{i+1}$ for all values of i and mean square deviation of the group is minimum about the number 'a_{51}', then mode of the group will be -

(A) $2a_{51}$ (B) a_{51} (C) a_{50} (D) a_{52}

14. If mean deviations about median of x, 2x, 3x, 4x, 5x, 6x, 7x, 8x, 9x, 10x is 30, then $|x|$ equals -

(A) 12 (B) 11 (C) 10 (D) 9

15. Mean and variance of a set of 6 terms is 11 and 24 respectively and the mean and variance of another set of 3 terms is 14 and 36 respectively. Then variance of all 9 terms is equal to -

(A) 40 (B) 30 (C) 50 (D) 35

16. Let $x_1, x_2, \ldots\ldots\ldots, x_{100}$ are 100 observations such that $\sum x_i = 0$, $\sum_{1\le i<j\le 100} |x_i x_j| = 80000$ & mean deviation from their mean is 5, then their standard deviation, is-

(A) 10 (B) 30 (C) 40 (D) 50

17. The A.M. of a set of 50 numbers is 38. If two numbers of the set namely 55 and 45 are discarded, the A.M. of the remaining set of numbers is

(A) 36 (B) 36.5 (C) 37.5 (D) 38.5

18. Let $x_1, x_2 \ldots\ldots x_n$ be n observations such that $\Sigma x_i^2 = 200$ and $\Sigma x_i = 40$ then a possible value of n among the following is

(A) 4 (B) 5 (C) 6 (D) 10

19. The arithmetic mean of a set of observation is $\bar{X}$. If each observation is divided by α and then is increased by 10, then the mean of the new series is

(A) $\frac{\bar{X}}{\alpha}$ (B) $\frac{\bar{X}+10}{\alpha}$ (C) $\frac{\bar{X}+10\alpha}{\alpha}$ (D) $\alpha\bar{X}+10$

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SECTION–II

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. A relation R is defined on set A = {a,b,c} such that R ≡ {(a,a), (b,b),(b,c),(c,b)}, then minimum number of additional elements required in R so that R becomes an equivalance relation, is

2. Let the mean and variance of four numbers 3, 7, x and y(x > y) be 5 and 10 respectively. Then the mean of four numbers 3 + 2x, 7 + 2y, x + y and x – y is _____.

3. Let $X_1, X_2, \ldots, X_{18}$ be eighteen observations such that $\sum_{i=1}^{18}(X_i-\alpha)=36$ and $\sum_{i=1}^{18}(X_i-\beta)^2=90$, where α and β are distinct real numbers. If the standard deviation of these observations is 1, then the value of $|\alpha-\beta|$ is _____.

4. The variance of 10 observations is 16. If each observation is doubled, then standard deviation of new data will be -

5. In a given frequency distribution, the respective values of mean and median are 21 and 22. The value of mode is

6. If mean deviation of sample 1, 2, 3, x about 3 is 1 (where x > 3), then possible value of variance is

REVISION ASSINGMENT # 19 (LOG, MOD, TRI AND TE) MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	C	B	D	A	B	B	C	C
	Q.	11	12	13	14	15	16				
	A.	A,B,C	A,C	A,C	A,D	B,D	A,B,C				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	7	0	7	3	4	6				

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